AUTOMATED PLANETARY PERCEPTION SYSTEM

Part 15: Factor Graph Estimation & 3D Epistemic Mapping

Level 3, Volume 3 of 5: iSAM2 Smoothing, IMU Preintegration, Multi-Resolution OctoMap & ESDF Fields

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VOLUME SUMMARY: *Specifies factor graph SLAM state estimation (iSAM2 smoothing on SO(3) x R^6), IMU preintegration, stereo visual odometry, slip-compensated wheel odometry, multi-resolution OctoMap representation, and Euclidean Signed Distance Fields (ESDF).*

Chapter 260. Executive Summary & State Estimation Framework

Context & Operational Boundaries: Defines the Level 3 factor graph state estimation and continuous 3D geometric/epistemic mapping architecture.

• **Factor Graph Smoothing:** Employs iSAM2 incremental smoothing over manifold state variables including chassis pose, velocity, and sensor biases.

• **Multi-Resolution Mapping:** Octree-based 3D occupancy and epistemic uncertainty mapping with continuous Euclidean distance transforms.

• **Drift Propagation:** Localisation uncertainty covariance Sigma_pose is mathematically propagated into obstacle buffer inflations.

	Measurement Source	Error Metric	Covariance Weighting
on	Tactical IMU (200 Hz)	Manifold Delta R, Delta v, Delta p Residuals	Continuous IMU Noise Covariance
	Stereo Navcam Pairs (5 Hz)	Reprojection Feature Error	Stereo Disparity Covariance
	Drive Encoders (50 Hz)	Kinematic Arc Displacements	Terramechanical Slip Penalty Covariance
	Global Feature Retrieval	SE(3) Relative Pose Error	Appearance Match Confidence Weight

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Chapter 260. Chapter 260.1: Factor Graph & Mapping Specification 1

Level 3 factor graph optimization and 3D mapping specification page 4.

Equation (264.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 4.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 260. Chapter 260.2: Factor Graph & Mapping Specification 2

Level 3 factor graph optimization and 3D mapping specification page 5.

Equation (265.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 5.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 260. Chapter 260.3: Factor Graph & Mapping Specification 3

Level 3 factor graph optimization and 3D mapping specification page 6.

Equation (266.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 6.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 261. Chapter 261.1: Factor Graph & Mapping Specification 4

Level 3 factor graph optimization and 3D mapping specification page 7.

Equation (267.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 7.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 261. Chapter 261.2: Factor Graph & Mapping Specification 5

Level 3 factor graph optimization and 3D mapping specification page 8.

Equation (268.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 8.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 261. Chapter 261.3: Factor Graph & Mapping Specification 6

Level 3 factor graph optimization and 3D mapping specification page 9.

Equation (269.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 9.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 262. Chapter 262.1: Factor Graph & Mapping Specification 7

Level 3 factor graph optimization and 3D mapping specification page 10.

Equation (270.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 10.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 262. Chapter 262.2: Factor Graph & Mapping Specification 8

Level 3 factor graph optimization and 3D mapping specification page 11.

Equation (271.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 11.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 262. Chapter 262.3: Factor Graph & Mapping Specification 9

Level 3 factor graph optimization and 3D mapping specification page 12.

Equation (272.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 12.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 263. Chapter 263.1: Factor Graph & Mapping Specification 10

Level 3 factor graph optimization and 3D mapping specification page 13.

Equation (273.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 13.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 263. Chapter 263.2: Factor Graph & Mapping Specification 11

Level 3 factor graph optimization and 3D mapping specification page 14.

Equation (274.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 14.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 263. Chapter 263.3: Factor Graph & Mapping Specification 12

Level 3 factor graph optimization and 3D mapping specification page 15.

Equation (275.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 15.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 264. Chapter 264.1: Factor Graph & Mapping Specification 13

Level 3 factor graph optimization and 3D mapping specification page 16.

Equation (276.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 16.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 264. Chapter 264.2: Factor Graph & Mapping Specification 14

Level 3 factor graph optimization and 3D mapping specification page 17.

Equation (277.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 17.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 264. Chapter 264.3: Factor Graph & Mapping Specification 15

Level 3 factor graph optimization and 3D mapping specification page 18.

Equation (278.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 18.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 265. Chapter 265.1: Factor Graph & Mapping Specification 16

Level 3 factor graph optimization and 3D mapping specification page 19.

Equation (279.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 19.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 265. Chapter 265.2: Factor Graph & Mapping Specification 17

Level 3 factor graph optimization and 3D mapping specification page 20.

Equation (280.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 20.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 265. Chapter 265.3: Factor Graph & Mapping Specification 18

Level 3 factor graph optimization and 3D mapping specification page 21.

Equation (281.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 21.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 266. Chapter 266.1: Factor Graph & Mapping Specification 19

Level 3 factor graph optimization and 3D mapping specification page 22.

Equation (282.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 22.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 266. Chapter 266.2: Factor Graph & Mapping Specification 20

Level 3 factor graph optimization and 3D mapping specification page 23.

Equation (283.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 23.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 266. Chapter 266.3: Factor Graph & Mapping Specification 21

Level 3 factor graph optimization and 3D mapping specification page 24.

Equation (284.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 24.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 267. Chapter 267.1: Factor Graph & Mapping Specification 22

Level 3 factor graph optimization and 3D mapping specification page 25.

Equation (285.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 25.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 267. Chapter 267.2: Factor Graph & Mapping Specification 23

Level 3 factor graph optimization and 3D mapping specification page 26.

Equation (286.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 26.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 267. Chapter 267.3: Factor Graph & Mapping Specification 24

Level 3 factor graph optimization and 3D mapping specification page 27.

Equation (287.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 27.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 268. Chapter 268.1: Factor Graph & Mapping Specification 25

Level 3 factor graph optimization and 3D mapping specification page 28.

Equation (288.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 28.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 268. Chapter 268.2: Factor Graph & Mapping Specification 26

Level 3 factor graph optimization and 3D mapping specification page 29.

Equation (289.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 29.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 268. Chapter 268.3: Factor Graph & Mapping Specification 27

Level 3 factor graph optimization and 3D mapping specification page 30.

Equation (290.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 30.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 269. Chapter 269.1: Factor Graph & Mapping Specification 28

Level 3 factor graph optimization and 3D mapping specification page 31.

Equation (291.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 31.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 269. Chapter 269.2: Factor Graph & Mapping Specification 29

Level 3 factor graph optimization and 3D mapping specification page 32.

Equation (292.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 32.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 269. Chapter 269.3: Factor Graph & Mapping Specification 30

Level 3 factor graph optimization and 3D mapping specification page 33.

Equation (293.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 33.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 270. Chapter 270.1: Factor Graph & Mapping Specification 31

Level 3 factor graph optimization and 3D mapping specification page 34.

Equation (294.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 34.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 270. Chapter 270.2: Factor Graph & Mapping Specification 32

Level 3 factor graph optimization and 3D mapping specification page 35.

Equation (295.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 35.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 270. Chapter 270.3: Factor Graph & Mapping Specification 33

Level 3 factor graph optimization and 3D mapping specification page 36.

Equation (296.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 36.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit

Chapter 271. Chapter 271.1: Factor Graph & Mapping Specification 34

Level 3 factor graph optimization and 3D mapping specification page 37.

Equation (297.1): Factor Graph Cost Optimization & Distance Field

```
X_opt = argmin_X sum_k || r_k( X_k, z_k ) ||_{Sigma_k}^2 ESDF(x) = min_{p in Occupied} || x - p ||_2 - min_{q in Free} || x - q ||_2
```

Where: X is rover trajectory state, r_k is residual vector for factor k, Sigma_k is measurement noise covariance, ESDF(x) is signed obstacle distance.

- **State Variable Optimization:** Solves sparse Levenberg-Marquardt steps with incremental Cholesky factorization in sector 37.
- **Dynamic Voxel Clearing:** Requires N_clear = 3 consecutive multi-frame observations before clearing confirmed obstacles.

Mapping Metric	Target Resolution	Memory Allocation
Octree Voxel Size	0.05 m	128 MB Bounded Cache
ESDF Horizon	15.0 m Local Radius	Ring-Buffer GPU/CPU Shared
Max Covariance Trace	0.25 m^2 (X/Y)	Safe Stopping Limit